



Statistics of Financial Markets

Seminar 4: Probability



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CDF: Cumulative Distribution Function
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ LCG: Linear Congruential Generator
- ▣ PCG64: Permuted Congruential Generator, 64-bit (NumPy default)
- ▣ PMF: Probability Mass Function
- ▣ RANDU: RANDU (IBM random number generator of the 1960s)

Today's Question and Route

- **Question:** how do we describe, condition and simulate random returns and prices?
 - ▶ this seminar comes **before** Lecture 4: the section "What You Need for Today" gives every definition the tasks use
- Route
 - ▶ Part A: moments, joint tables, conditioning, binomial trees, AR(1), inverse transform and GBM on paper
 - ▶ Part B: dependence in real returns, Monte Carlo, random number generators and GBM against data, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- Nothing is handed in: the seminar is for practice; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	moments of a discrete return; is an uncorrelated pair independent?	Solved, Proposed	A1
A3, A4	portfolio volatility; a two-regime model and the law of total variance	Solved, Proposed	A3
A5, A6	a binomial tree; random walk, martingale and AR(1)	Solved, Proposed	A5
A7, A8	inverse transform by hand; GBM and the Wiener process	Solved, Proposed	A7
B1, B2	are returns independent? conditional mean and variance	Solved, Proposed	B1
B3, B4	a loss probability by Monte Carlo; testing a random number generator	Solved, Proposed	B3
B5, B6	GBM against the BET; does variance grow linearly with the horizon?	Solved, Proposed	B5
C1, C2	how often is there a bad year? what is wrong in an AI answer?	Proposed	B3, B5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Weekends and repeated holiday closes are dropped (except Bitcoin); each series on its own calendar
- Simulations use NumPy's default generator with fixed seeds, so every number can be reproduced

What You Need for Today (1/4): Random Variables and Moments

- Discrete X : **PMF** (probability mass function) $p(x_i) = P(X = x_i)$; continuous X : **PDF** (probability density function) f
 - ▶ **CDF** (cumulative distribution function): $F(x) = P(X \leq x)$; quantile $q_\alpha = F^{-1}(\alpha)$
- $E[X] = \sum x_i p(x_i)$ or $\int x f(x) dx$; $\text{Var}(X) = E[X^2] - (E[X])^2$
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$; $\rho = \text{Cov}/(\sigma_X \sigma_Y)$; $\text{Var}(aX + bY) = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \text{Cov}$
- **Independent**: $P(X = x, Y = y) = P(X = x)P(Y = y)$ for every cell; independent $\Rightarrow$ uncorrelated, not conversely
 - ▶ the i.i.d. band for a sample correlation: $\pm 1.96/\sqrt{n}$

What You Need for Today (2/4): Conditioning

- $P(A | B) = P(A \cap B) / P(B)$; $E[Y | X]$: the mean of Y once X is known, the best forecast of Y from X
- **Tower property:** $E[E[Y | X]] = E[Y]$
- **Law of total variance:** $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$
- For returns: $\mu_t = E[r_t | \mathcal{F}_{t-1}]$ and $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information up to yesterday
 - ▶ a changing σ_t is what GARCH models ([Chapter 9](#))
- A mixture of Normals with different variances has kurtosis above 3: $E[X^4] = 3\sigma^4$ for $N(0, \sigma^2)$

What You Need for Today (3/4): Processes

- **White noise:** mean 0, variance σ^2 , uncorrelated over time; **i.i.d. noise:** independent and identically distributed
- **Random walk:** $S_t = S_{t-1} + \varepsilon_t$, $E[S_t] = S_0$, $\text{Var}(S_t) = t\sigma^2$
- **Martingale:** $E[X_{t+1} | \mathcal{F}_t] = X_t$ (a fair game)
- **AR(1):** $X_t = c + \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$: mean $c/(1 - \phi)$, variance $\sigma^2/(1 - \phi^2)$, $\rho(h) = \phi^h$
- **Binomial tree:** up by u with probability p , down by d ; after n steps $S_n = S_0 u^K d^{n-K}$, $K \sim B(n, p)$
 - ▶ $E[S_n] = S_0(pu + (1 - p)d)^n$; fair game for $p^* = (1 - d)/(u - d)$

What You Need for Today (4/4): Simulation, Wiener and GBM

- ▣ **Inverse transform:** $U \sim U(0, 1) \Rightarrow F^{-1}(U)$ has CDF F ; exponential: $-\ln(1 - U)/\lambda$; Normal: $\mu + \sigma\Phi^{-1}(U)$
- ▣ **LCG** (linear congruential generator): $x_{k+1} = (ax_k + c) \bmod M$, $u_k = x_k/M$
- ▣ **Monte Carlo:** $\hat{p}$ = share of simulations in the event; standard error $\sqrt{p(1-p)/N}$
- ▣ **Wiener process:** $W(0) = 0$, independent increments, $W(t) - W(s) \sim N(0, t - s)$; $\text{Cov}(W(s), W(t)) = \min(s, t)$
- ▣ **GBM** (geometric Brownian motion): $S_T = S_0 \exp((\mu - \sigma^2/2)T + \sigma W_T)$
 - ▶ mean $S_0 e^{\mu T}$, median $S_0 e^{(\mu - \sigma^2/2)T}$; daily log returns i.i.d. Normal
 - ▶ maximum drawdown: $\min_t(P_t / \max_{s \leq t} P_s - 1)$, the largest fall from a running peak

A1: Moments of a Discrete Return [Solved]

Task

- A stock's daily return X (in %) takes the values -3 , 0 and 2 with probabilities 0.2 , 0.3 and 0.5 .
- 1. Compute $E[X]$, $E[X^2]$, $\text{Var}(X)$ and the standard deviation.
- 2. Write the CDF $F(x)$ for every x .
- 3. Compute $P(X < 0)$ and the median.
- Report: four moments, the CDF and two numbers.

Solution

- 1. $E[X] = -0.6 + 0 + 1 = 0.40$; $E[X^2] = 1.8 + 2 = 3.80$;
 $\text{Var} = 3.80 - 0.40^2 = 3.64$; s.d. 1.908
- 2. $F(x) = 0$ for $x < -3$; 0.2 on $[-3, 0)$; 0.5 on $[0, 2)$; 1 for $x \geq 2$
- 3. $P(X < 0) = 0.2$; median $= \inf\{x : F(x) \geq 0.5\} = 0$
- The mean is positive although a loss is possible: risk and expected return are different questions.

A2: Uncorrelated but Dependent [Proposed]

Solution discussed in class

Task

- ▣ X = market move (-1 fall, 0 flat, $+1$ rise); $Y = 1$ if the day is "large". Joint probabilities: $P(-1, 1) = 0.25$, $P(0, 0) = 0.5$, $P(1, 1) = 0.25$, all other cells 0 . Model: A1.
- ▣ 1. Compute the marginal distributions of X and Y and their means.
- ▣ 2. Compute $\text{Cov}(X, Y)$ and ρ .
- ▣ 3. Decide whether X and Y are independent, using one cell of the table.
- ▣ 4. Compute $\text{Var}(X + Y)$.
- ▣ Report: two marginals, ρ , a verdict and a variance.

A3: Volatility of a Two-Asset Portfolio [Solved]

Task

- ▣ Two assets with annual volatilities $\sigma_1 = 20\%$ and $\sigma_2 = 30\%$; weights 0.6 and 0.4.
- ▣ 1. Compute the portfolio volatility for $\rho = 0.3$.
- ▣ 2. Repeat for $\rho = 0$, $\rho = -0.5$ and $\rho = 1$.
- ▣ 3. Explain when the portfolio is less risky than the weighted average of the two volatilities.
- ▣ Report: four volatilities and one sentence.

Solution

- ▣ 1.
 $\sigma_p^2 = 0.36 \times 400 + 0.16 \times 900 + 2 \times 0.24 \times 0.3 \times 600 = 374.4$;
 $\sigma_p = 19.35\%$
- ▣ 2. $\rho = 0$: 16.97%; $\rho = -0.5$: 12.00%; $\rho = 1$: 24.00%
- ▣ 3. Only $\rho = 1$ gives the weighted average
 $0.6 \times 20 + 0.4 \times 30 = 24\%$; every $\rho < 1$ diversifies.

A4: Two Regimes and the Law of Total Variance [Proposed]

Solution discussed in class

Task

- A day is calm with probability 0.8, $r \mid \text{calm} \sim N(0, 0.8^2)$, and turbulent otherwise, $r \mid \text{turbulent} \sim N(0, 2^2)$ (in %). Model: A3.
- 1. Use the law of total variance to compute $\text{Var}(r)$ and the standard deviation.
- 2. Compute $E[r^4]$ with $E[Z^4] = 3\sigma^4$ in each regime, then the kurtosis $E[r^4]/\text{Var}(r)^2$.
- 3. Explain why the mixture has heavy tails although each regime is Normal.
- Report: a variance, a kurtosis and one sentence.

A5: A Three-Step Binomial Tree [Solved]

Task

- $S_0 = 100$, $u = 1.1$, $d = 0.9$, $p = 0.55$, three independent steps.
- 1. List the possible values of S_3 and their probabilities.
- 2. Compute $E[S_3]$ in two ways: from the table and from $S_0(pu + (1 - p)d)^3$.
- 3. Compute $P(S_3 < 100)$.
- 4. Find the p^* that makes S_t a martingale.
- Report: a table, two expectations, a probability, p^* .

Solution

- 1. $K = 0..3$ up moves: 72.9, 89.1, 108.9, 133.1 with 0.0911, 0.3341, 0.4084, 0.1664 ($\binom{3}{k} p^k (1 - p)^{3-k}$)
- 2. $E[S_3] = 100 \times 1.01^3 = 103.03$ both ways
- 3. $P(K \leq 1) = 0.4253$
- 4. $p^* = (1 - 0.9)/(1.1 - 0.9) = 0.50$: with $p = 0.55 > p^*$ the price drifts up

A6: Random Walk, Martingale and AR(1) [Proposed]

Solution discussed in class

Task

- ▣ $S_t = S_{t-1} + \varepsilon_t$, ε_t i.i.d. with mean 0 and variance σ^2 ; $X_t = 0.2 + 0.6X_{t-1} + \varepsilon_t$ with $\sigma = 1$. Model: A5.
- ▣ 1. Compute $E[S_t]$ and $\text{Var}(S_t)$, and show that S_t is a martingale.
- ▣ 2. Show that $S_t^2 - t\sigma^2$ is also a martingale.
- ▣ 3. Compute the mean, the variance and $\rho(1)$, $\rho(2)$, $\rho(5)$ of the stationary X_t .
- ▣ 4. Compute $E[X_{t+1} \mid X_t = 2]$ and $\text{Var}(X_{t+1} \mid X_t = 2)$.
- ▣ Report: two derivations, five moments, a forecast.

A7: The Inverse Transform Step by Step [Solved]

Task

- You have the uniform numbers $u = 0.3$ and $u = 0.9$ (and $u = 0.15, 0.45, 0.8$ for question 2).
- 1. Turn $u = 0.3$ and $u = 0.9$ into exponential draws with rate $\lambda = 0.5$.
- 2. Turn $u = 0.15, 0.45, 0.8$ into draws of the return X of A1.
- 3. Turn $u = 0.01$ into a draw of $N(0.05, 1.2^2)$.
- Report: six simulated values.

Solution

- 1. $x = -\ln(1 - u)/0.5$: 0.713 and 4.605
- 2. $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ with $F = 0.2, 0.5, 1$: -3, 0 and 2
- 3. $0.05 + 1.2 \times (-2.326) = -2.742$: the 1% quantile
- A large u gives a large draw: the inverse transform keeps the order of the uniforms.

A8: GBM and the Wiener Process [Proposed]

Solution discussed in class

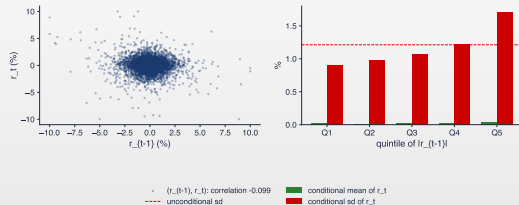
Task

- ▣ An index follows GBM with $\mu = 8\%$, $\sigma = 20\%$ a year, $S_0 = 100$; W is a Wiener process. Model: A7.
- ▣ 1. Compute the mean and the median of S_1 and of S_5 .
- ▣ 2. Compute $P(S_T < 100)$ for $T = 1$ and $T = 5$.
- ▣ 3. Compute the 5% quantile of S_1 and of S_5 .
- ▣ 4. Compute $\text{Var}(W(2) - W(0.5))$ and $\text{Cov}(W(1), W(3))$.
- ▣ Report: eight GBM numbers and two Wiener moments.

B1: Are S&P 500 Returns Independent? [Solved]

- ▣ **Question:** are the daily returns of the S&P 500 independent, and what does yesterday tell us about today?
- ▣ **Inputs:** S&P 500 closes, 2000–2026, daily log returns in %
- ▣ **Tasks**
 1. Compute $\text{Corr}(r_t, r_{t-1})$ and $\text{Corr}(r_t^2, r_{t-1}^2)$ and compare them with the band $\pm 1.96/\sqrt{n}$.
 2. Split the days into quintiles of $|r_{t-1}|$ and compute the mean and the standard deviation of r_t in each quintile.
 3. Check the law of total variance with the quintiles as groups.
 4. Interpretation: which is predictable from yesterday, the mean or the variance of today's return?
- ▣ **Report:** two correlations, a table of five quintiles, one decomposition, one sentence
- ▣ **Notebook:** section B1

B1: Solution [Solved]



- $n = 6\,714$, band ± 0.024 ; $\text{Corr}(r_t, r_{t-1}) = -0.099$;
 $\text{Corr}(r_t^2, r_{t-1}^2) = 0.313$
- Conditional means Q1–Q5: 0.021, 0.006, 0.029, 0.027, 0.042 (%)
- Conditional s.d.: 0.91, 0.98, 1.07, 1.21, 1.71; $Q5/Q1 = 1.88$
- Total variance $1.4691 = 1.4690$ (within) $+0.00013$ (between, 0.01%)
- Interpretation: the variance is predictable, the mean almost not; the negative lag-1 correlation comes from crisis days with large reversals, so returns are not independent

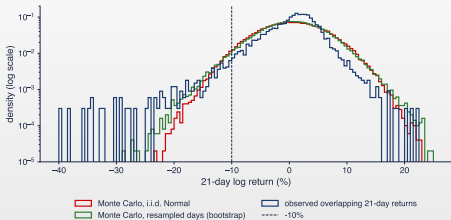
B2: The Same for DAX, BET and Bitcoin [Proposed]

- ▣ **Question:** do the DAX, the BET and Bitcoin show the same pattern as the S&P 500?
- ▣ **Inputs:** daily log returns, 2000–2026 (Bitcoin from 2014); model: B1
- ▣ **Tasks**
 1. Compute both lag-1 correlations and the band for each series.
 2. Compute the conditional standard deviation in Q1 and Q5 and their ratio.
 3. Compute the share of the variance explained by the quintile means.
 4. Interpretation: why is the BET the only series with a clearly positive $\text{Corr}(r_t, r_{t-1})$?
- ▣ **Report:** one table and one sentence
- ▣ **Notebook:** section B2

B3: A Monthly Loss Probability by Monte Carlo [Solved]

- ▣ **Question:** how likely is a 21-day loss of more than 10% for the S&P 500, and does the answer depend on the model?
- ▣ **Inputs:** S&P 500 daily log returns, 2000–2026; 100 000 simulated months
- ▣ **Tasks**
 1. Compute the exact probability under i.i.d. Normal daily returns with the sample mean and standard deviation.
 2. Estimate it by Monte Carlo under the same model and give the standard error.
 3. Estimate it by resampling 21 observed days with replacement (the empirical inverse transform).
 4. Compute the share of overlapping 21-day windows in the data with a loss beyond 10%.
 5. Interpretation: which differences are simulation error and which are model error?
- ▣ **Report:** four probabilities, two standard errors, one sentence
- ▣ **Notebook:** section B3

B3: Solution [Solved]



- Exact Normal: 2.922%; Monte Carlo Normal: 2.917% (s.e. 0.053%); bootstrap: 3.184% (s.e. 0.056%)
- Data: 2.67% of 6694 overlapping windows (179 windows, from a few episodes); non-overlapping months: 3.45% of 319
- Interpretation: Monte Carlo vs exact differs by less than one s.e. (simulation error); bootstrap vs Normal differs by about 3.5 s.e. (model error: heavier daily tails); the data are noisy because overlapping windows are dependent

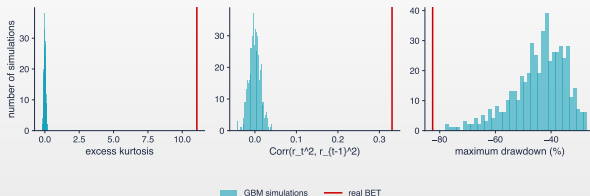
B4: Test a Random Number Generator [Proposed]

- **Question:** can a generator pass simple tests and still be unusable?
- **Inputs:** RANDU, $x_{k+1} = 65539 x_k \bmod 2^{31}$, seed 1, 100 000 numbers; NumPy PCG64 with the same seed; model: B3
- **Tasks**
 1. Compute the mean, the variance and the lag-1 correlation of both sequences and compare with $1/2$, $1/12$ and 0 .
 2. Count the distinct values of $9u_k - 6u_{k+1} + u_{k+2}$ for both generators.
 3. Find the period of the LCGs $(a, c, M) = (5, 1, 16)$, $(5, 0, 16)$ and $(4, 1, 16)$ started at 7 .
 4. Interpretation: why do one-dimensional tests not detect the flaw of RANDU?
- **Report:** a table of six statistics, two counts, three periods, one sentence
- **Notebook:** section B4

B5: Is the BET a Geometric Brownian Motion? [Solved]

- ▣ **Question:** which features of the BET can a GBM with the same drift and volatility reproduce?
- ▣ **Inputs:** BET closes, 2000–2026; 500 GBM paths of the same length
- ▣ **Tasks**
 1. Calibrate GBM: annual log drift and volatility from the daily log returns.
 2. Simulate 500 paths and compute, for each, the excess kurtosis, $\text{Corr}(r_t^2, r_{t-1}^2)$ and the maximum drawdown.
 3. Locate the real values in the simulated distributions (5% and 95% quantiles).
 4. Interpretation: which stylised facts does GBM miss, and which risk number is most affected?
- ▣ **Report:** two parameters, three comparisons, one sentence
- ▣ **Notebook:** section B5

B5: Solution [Solved]



- ▣ $n = 6\,670$ days; $\sigma = 22.1\%$, log drift 16.0% a year
- ▣ Excess kurtosis: real 11.06 , GBM 90% range $[-0.10, 0.10]$; $\text{Corr}(r_t^2, r_{t-1}^2)$: real 0.330 , GBM $[-0.022, 0.022]$
- ▣ Maximum drawdown: real -82.5% , GBM median -42.7% , range $[-62.8\%, -31.4\%]$
- ▣ Interpretation: GBM misses heavy tails and volatility clustering completely; the 2008 crash makes the real drawdown deeper than every simulated path, so a GBM-based risk limit would have been far too loose

B6: Does Variance Grow Linearly with the Horizon? [Proposed]

- **Question:** does $\text{Var}(h\text{-day return}) = h \text{Var}(1\text{-day return})$ hold, as a random walk with i.i.d. increments predicts?
- **Inputs:** S&P 500, BET, Bitcoin daily log returns; non-overlapping blocks of $h = 5, 21, 63$ days; model: B5
- **Tasks**
 1. Compute the ratio $\text{Var}(h\text{-day})/(h \text{Var}(1\text{-day}))$ for each series and horizon.
 2. Use the approximate standard error $\sqrt{2/(m-1)}$ of a variance ratio with m blocks to judge the distance from 1.
 3. Interpretation: what does a ratio above 1 or below 1 say about the autocorrelation of returns?
- **Report:** a 3×3 table and two sentences
- **Notebook:** section B6

C1: How Often Is There a Bad Year? [Proposed]

- **Question:** does GBM give the right probability of a drawdown beyond 20% (or 40%) within one calendar year?
- **Inputs:** S&P 500, DAX, BET, Bitcoin, complete calendar years 2000–2025 (Bitcoin 2015–2025); models: B3, B5
- **Tasks**
 1. Compute the maximum drawdown within each calendar year and the share of years beyond 20% and beyond 40%.
 2. Simulate 10 000 one-year paths under GBM and under the i.i.d. bootstrap of the daily returns and compute the same two probabilities.
 3. Compare data and models with a binomial test.
 4. Propose one model that could reproduce both numbers.
 5. Interpretation: what does the gap between the bootstrap and the data say about volatility clustering?
- **Report:** a table, two binomial p -values, a short project plan
- **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to “summarise the probability facts about the S&P 500 for a risk report”. The answer:
- (a) $\text{Corr}(r_t, r_{t-1}) = -0.099$, so daily returns are independent and tomorrow’s variance cannot be predicted.
- (b) The log price is a random walk, so its variance is the same at every date and the process is stationary.
- (c) Under GBM with $\mu = 8\%$ and $\sigma = 19\%$, the median value of 100 after 10 years is $100 \exp(0.8) = 222.6$.
- (d) A Monte Carlo probability of about 3% from 10,000 paths has a standard error of about 0.0017.
- (e) A 50/50 portfolio of S&P 500 and DAX has daily volatility $\sqrt{0.25 \times 1.22^2 + 0.25 \times 1.43^2} = 0.94\%$.
- **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of five verdicts with one line of justification each.

What You Should Take from Today

- ▣ Zero correlation does not mean independence: check squares or absolute values too
- ▣ The conditional variance of returns moves with yesterday's news; the conditional mean hardly moves
- ▣ Mixtures of Normal regimes give heavy tails
- ▣ Monte Carlo: report the standard error; separate simulation error from model error
- ▣ GBM reproduces the path of an index roughly, but not its tails, clusters or deepest drawdowns
- ▣ An AI answer is a draft: check every definition, model and number

After the Seminar

- Lecture 4 develops each topic of today: distributions, dependence, conditioning, Monte Carlo, binomial trees, random walks, Wiener process and GBM
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: GARCH and regime-switching simulations, block bootstrap, more indices
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 3–5; exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#)

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